

LAB 02: SNOW WHITE SPACE

Lab title	Snow white space
Lab goal / objective	Hide and reveal data in text file
Technical information	
Category	Steganography > White space

1.1. Lab brief introduction

Steganography, simply can be defined as “covered-writing” which was derived from Greek vocabulary. It hides the data and information with seamlessly innocent data. Johannes T. has used the “Steganography” term in 1499, in his book “Steganographia”.

Uses of Steganography

- Send information in secured manner
- Store information in secured manner
- As an extended implementation of water marking
- As an extended implementation of copyright protection

Types of Steganography

- Technical Steganography
- Linguistic Steganography
 - Semagrams
 - Text semagrams
 - Visual semagrams
 - Open codes
 - Jargon code
 - Covered ciphers
 - Null cipher
 - Grille cipher

Techniques of Steganography

- Network Steganography
- Audio Steganography
- Video Steganography
- Text Steganography
- Image Steganography

Impact of the Steganography

With the wrong intention, Steganography is a great threat to the security. It can be used to carry to conduct terrorist communication, illegal actions, crimes etc. Also, Steganography can be used as a medium to Trojans, to carry the malwares.

Risk mitigation / recommendation

- Use trusted sources to download resources
- Use well-secured and encrypted communication methods
- Do not click or download or open suspicious text/audio/image files from untrusted or unknown origins
- Use anti-malware tools to scan the resources
- Use sandboxing
- Be aware and be secured from phishing and social engineering

1.2. Lab setup

Following software are used.

Software	Description
Windows 10	The normal PC is used.
SNOW version 1.1	

1.3. Used tools

Snow - Version 1.1

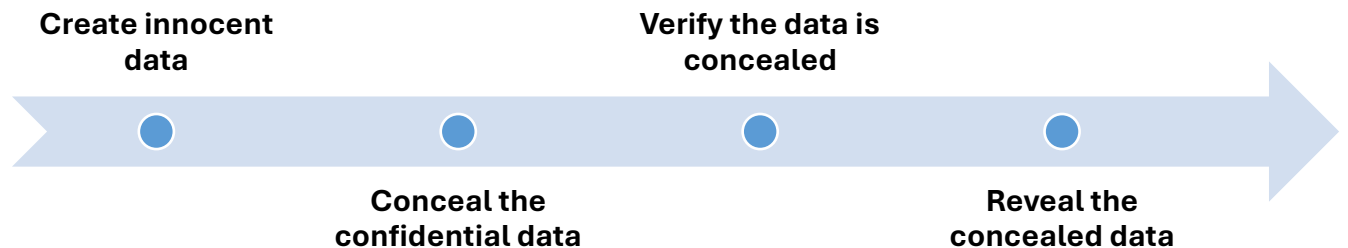
This tool, “SNOW” aka “Steganographic Nature Of Whitespace” allows users to hide the messages in ASCII format by simply extending the “whitespace” to the end-of-lines. SNOW utilize the Information Concealment Engine (ICE) encryption algorithm, a symmetric-key block cipher which was developed in 1997 by Mathew Kwan .The encryption and decryption is done through a password.

Command syntax:

```
snow [ -CQS ] [ -p passwd ] [ -l line-len ] [ -f file | -m message ] [ infile [ outfile ] ]
```

1.4. Lab implementation mapping and steps

1.4.1.Lab task flow



1.4.2.Lab implementation mapping with process flow

Task	Steps
Create innocent data	1 - 2
Conceal the confidential data	3 - 4
Verify the data is concealed	5
Reveal the concealed data	6

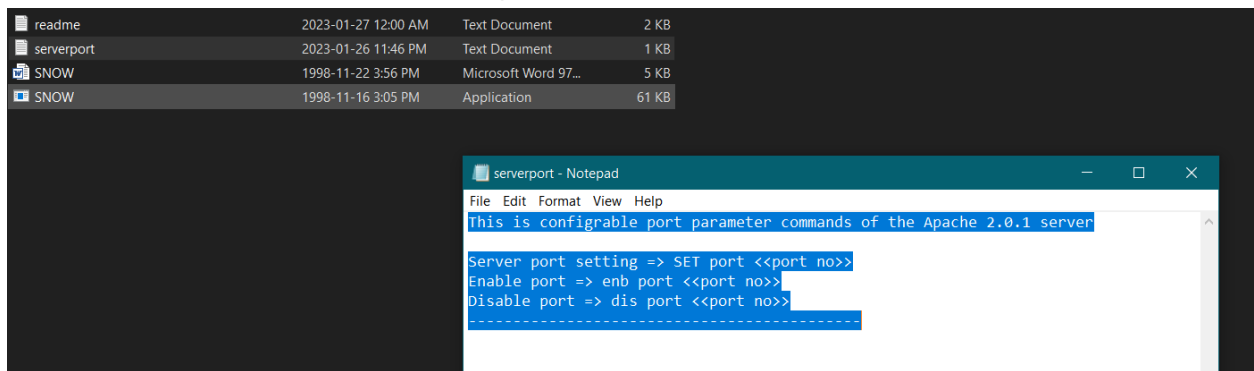
1. Navigate to the folder of the SNOW software and create the text file, the “innocent” data. Here it is **serverport.txt**

Name	Date modified	Type	Size
serverport	2023-01-26 11:46 PM	Text Document	1 KB
SNOW	1998-11-22 3:56 PM	Microsoft Word 97...	5 KB
SNOW	1998-11-16 3:05 PM	Application	61 KB

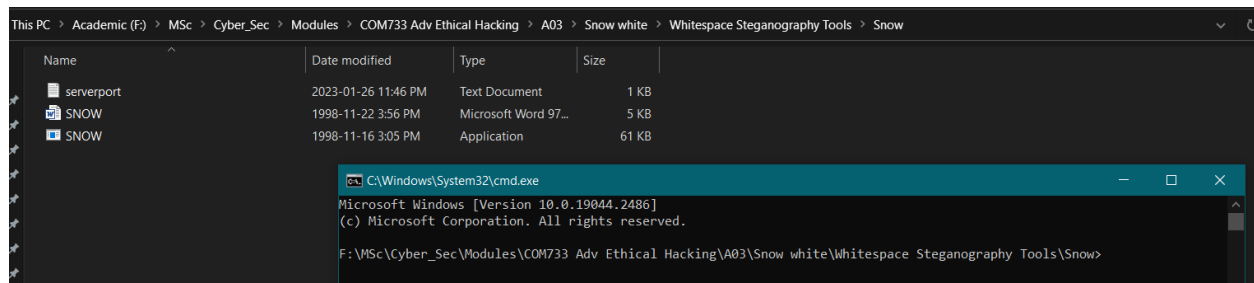
```
serverport.txt x new 2 x
1 This is configurable port parameter commands of the Apache 2.0.1 server
2
3 Server port setting => SET port <<port no>>
4 Enable port => enb port <<port no>>
5 Disable port => dis port <<port no>>
6 -----
```

2. Open the “serverport.txt” file and check the whitespaces are not available.

Edit -> Select All revealed the added whitespaces.



3. Open the command line in the folder where SNOW.exe is available



4. Conceal the data by following command.

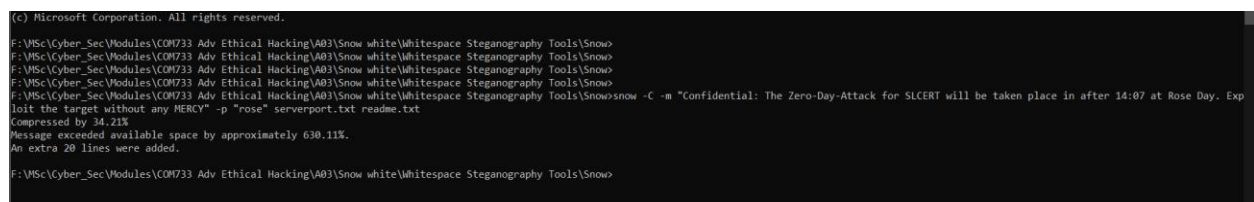
Confidential data: **Confidential: The Zero-Day-Attack for SLCERT will be taken place in after 14:07 at Rose Day. Exploit the target without any MERCY**

Password: **rose**

Input file: **serverport.txt**

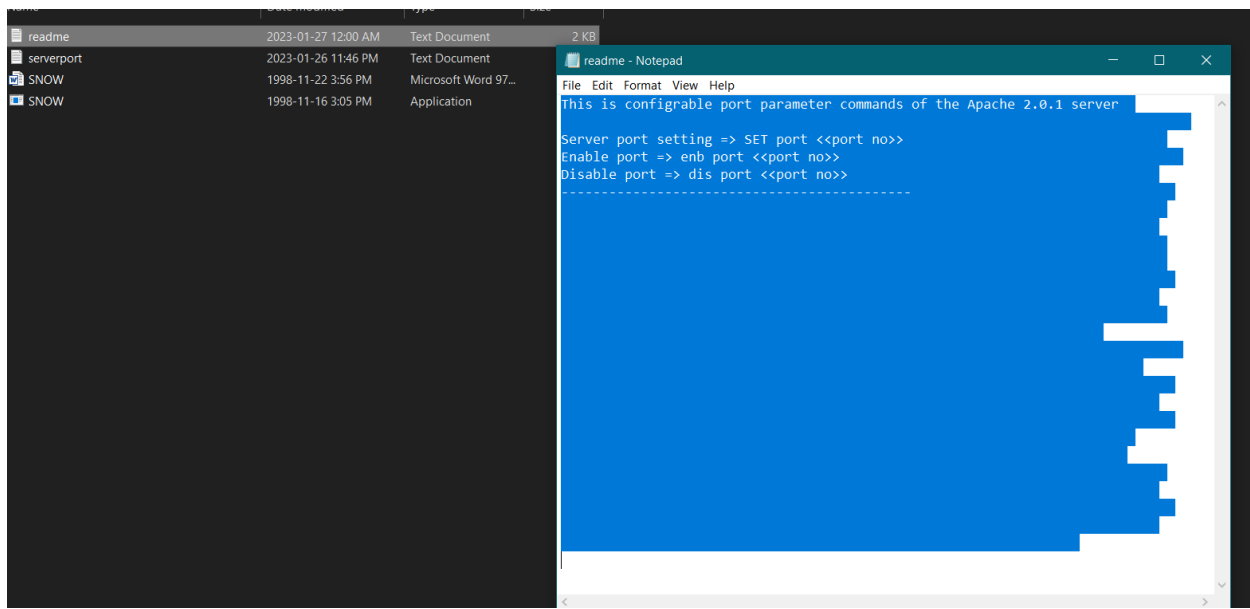
Output file: **readme.txt**

snow -C -m "Confidential: The Zero-Day-Attack for SLCERT will be taken place in after 14:07 at Rose Day. Exploit the target without any MERCY" -p "rose" serverport.txt readme.txt



5. Open the “serverport.txt” file and check the whitespaces are available.

Edit -> Select All revealed the added whitespaces.



6. Reveal the concealed data by following command.

```
snow -C -p "rose" readme.txt
```

